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CAPSTONE PROJECT REPORT: RETAIL SALES PERFORMANCE & CUSTOMER CHURN ANALYTICS DASHBOARD

Candidate Name:	Aman Gupta	Degree & Branch:	BCA (Computer Science)
College / University:	Poddar College, Bharatpur	Specialization:	Data Analytics
Industry Mentor:	Prof. Krishlay Sharma	Project Status:	APPROVED & DEFENDED

1. Problem Statement

An omnichannel retail chain operating across North India faced a 14% year-over-year dip in customer repeat purchase rate. The management lacked a consolidated real-time dashboard to track store performance and customer churn.

2. Enterprise Dataset Architecture

Relational retail database comprising 65,000+ orders, 12,000 customer profiles, and 3 years of transactional records.

3. Technology Stack & Analytical Tools

Microsoft Excel 365, PostgreSQL, Python (Pandas, NumPy), Power BI Desktop, DAX.

4. Core Analytical Objectives

- Build automated data ingestion pipeline.
- Design Star Schema in Power BI.
- Segment churn risk via RFM analysis.
- Deliver executive dashboards.

5. Methodology & Execution Framework

N/A

6. Data Cleaning & Sanitization Protocols

N/A

7. Exploratory Data Analysis & Statistical Findings

N/A

8. Dashboard Visualization & Reporting

N/A

9. Executive Findings & Business Insights

Top 20% loyal customers drove 64% of total revenue. Customers inactive for 45 days had 82% churn likelihood.

10. Strategic Recommendations & ROI

Deploy automated Day-30 SMS/Email re-engagement triggers to recover estimated Rs. 42 Lakhs annually.

11. Conclusion & Operational Impact

Delivered robust BI solution reducing reporting turnaround from 5 days to real-time.

STUDENT SIGNATURE	FACULTY / MENTOR	AUTHORIZED SIGNATORY
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Candidate's Signature

Prof. Krishlay Sharma
Professor

Nitin Sir
Centre Head & Authorized Signatory
(Official Stamp & Seal)